

LogBase with a general tail rate:

$$|G(L) - A| \leq ML^{-\varepsilon} \implies |H_m(L) - \sum c_{m,i} \log^i L| \leq (M/\varepsilon^m) L^{-\varepsilon}$$

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Abstract

Unit 49's base predicate required the tail $|G(L) - A| \leq M/L$. The mixed-multiplicity case needs bases whose tail decays only like $L^{-(q-p)}$, where $q > p$ are the non-minimal and minimal candidate exponents. This unit generalises **LogBase** by a rate $\varepsilon > 0$ and shows that the transfer step preserves the rate at the cost of a factor $1/\varepsilon$ per level: $|\theta_m(L)| \leq (M/\varepsilon^m) L^{-\varepsilon}$. The weighted primitives are the case $\varepsilon = 1$, and every downstream statement is unchanged. Zero sorry/axiom.

1 Setting

With $|\theta_m(x)| \leq M'x^{-\varepsilon}$ the tail integrand satisfies $|\theta_m(x)/x| \leq M'x^{-\varepsilon-1}$, integrable on $[1, \infty)$ for any $\varepsilon > 0$, and $\int_L^\infty M'x^{-\varepsilon-1} = M'L^{-\varepsilon}/\varepsilon$. So the recursion $\theta_{m+1}(L) = -\int_L^\infty \theta_m/x$ yields $|\theta_{m+1}(L)| \leq (M'/\varepsilon)L^{-\varepsilon}$.

2 The things formalised

```
structure LogBase (G :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (A M C :  $\mathbb{R}$ ) : Prop where
...
_pos : 0 <
tail :  $L, 1 \leq L \rightarrow |G L - A| \leq M * L^{-\_pos}$ 
theorem divRem_tail_abs_le (G :  $\mathbb{R} \rightarrow \mathbb{R}$ ) (A M :  $\mathbb{R}$ ) (h : 0 <  $\_pos$ ) (m :  $\mathbb{N}$ ) (L :  $\mathbb{R}$ ) (hL : 1  $\leq$  L) :
  (hb :  $L, 1 \leq L \rightarrow |\text{divRem } G A m L| \leq M * L^{-\_pos}$ ) :
  | x in Ioi L, divRem G A m x / x|  $\leq$  M /  $\_pos^m * L^{-\_pos}$ 
theorem divRem_abs_le (hG : LogBase G A M C  $\_pos$ ) (m :  $\mathbb{N}$ ) :
   $L, 1 \leq L \rightarrow |\text{divRem } G A m L| \leq M / \_pos^m * L^{-\_pos}$ 
theorem iterDivPrim_expansion (hG : LogBase G A M C  $\_pos$ ) (m :  $\mathbb{N}$ ) (L :  $\mathbb{R}$ ) (hL : 1  $\leq$  L) :
  |iterDivPrim G m L - i|  $\leq$  Finset.range (m + 1), divCoeff G A m i * Real.log L  $^i$  |
  M /  $\_pos^m * L^{-\_pos}$ 
```

3 Proof strategy

A mechanical generalisation: the exponent -2 becomes $-\varepsilon - 1$ in the domination and in `integral_Ioi_rpow_of_lt`, the pointwise bound uses `Real.rpow_sub_one`, and the induction carries M/ε^m via `pow_succ` and `div_div`. In the asymptotic of unit 50 the remainder is bounded by M/ε^m on $L \geq 1$ using `Real.rpow_le_one_of_one_le_of_nonpos`. The F_γ instance sets $\varepsilon = 1$ and rewrites L^{-1} as $1/L$ so the previously stated corollary keeps its form.

4 Roadblocks and resolutions

- None: both files compiled on the first pass after the edit. The only care was to thread the level-dependent constant M' through the lemmas that take the remainder bound as a hypothesis.

5 What was Mathlib, what was new

Mathlib: `Real.rpow_sub_one`, `integral_Ioi_rpow_of_lt`, and the bound `Real.rpow_le_one_of_one_le_of_nonpos` for $L^{-\varepsilon} \leq 1$. New: the rate parameter and its propagation.

6 Lessons

When an abstraction is introduced (unit 49), the first consumer beyond the motivating instance usually exposes a hypothesis that was accidentally too strong; here the fix was a five-minute generalisation because the abstraction had isolated it.

7 Outcome

The iterated expansion is available for bases with any polynomial tail rate. Next: the mixed case with one noncritical coordinate innermost, whose base $G_1(y) = \int_0^y x^{p-q-1} F_{q-1}$ is a **LogBase** with $A = C(p, q)$, $\varepsilon = q - p$, giving $Z \sim c n^{-p/2} (\log n)^{m-1}$ with m the number of minimal coordinates.